

Matrices & Vectors



Addition and scalar multiplication

- 1 Let A be the matrix with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and B be the matrix with rows $\langle 2, -1 \rangle$ and $\langle 0, 5 \rangle$. Find $A + B$ and $2A$.
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Matrix multiplication

- 2 Using A (rows $\langle 1, 3 \rangle, \langle -2, 4 \rangle$) and B (rows $\langle 2, -1 \rangle, \langle 0, 5 \rangle$) from above, find AB .
 - 3 Let A have rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and let $\mathbf{v} = \langle 3, -1 \rangle$. Find $A\mathbf{v}$.
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Determinant and inverse

- 4 Find $\det A$ for the matrix A with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and state whether A is invertible.
 - 5 Find A^{-1} for the matrix A with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$.
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Solving systems with the inverse matrix

- 6 Solve the system $x + 3y = 13$ and $-2x + 4y = 4$ using $\mathbf{x} = A^{-1}\mathbf{b}$ (using A^{-1} from above).
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Vector magnitude and dot product

- 7 For $\mathbf{u} = \langle 4, 3 \rangle$ and $\mathbf{w} = \langle -3, 4 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.
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Angle between vectors

- 8 Find the angle between $\mathbf{p} = \langle 2, 2 \rangle$ and $\mathbf{q} = \langle 3, 0 \rangle$.
- 9 Find the angle between $\langle 7, 1 \rangle$ and $\langle 2, -3 \rangle$, rounded to the nearest tenth of a degree.